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Fluency oriented intervention for struggling readers: reading fluency gains and neural changes in visual word processing

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Introduction

Dyslexia is a disorder in the neural network for reading, with dysfluent reading as its most persistent symptom (Gabrieli, 2009). Reading ability and disability are not strictly categorisable, but are presented on a continuum, in which good and poor readers keep their relative position over time (Shaywitz & Shaywitz, 2005). Struggling readers, not diagnosed with dyslexia, display difficulties in transition from the decoding to the fluent word recognition(Kuhn et al., 2006), while dyslexics may exhibit difficulties already at the letter-speech sound integration level (Blomert, 2011).

Neuroimaging studies suggest a Visual Word Form Area (VWFA) in left occipito-temporal regions specialized for print reading (McCandliss, Cohen, & Dehaene, 2003). Electrophysiological data suggest that early N170 responses at around 200 ms are sensitive to word-likeness of stimuli and reading expertise (Maurer et al.,2007).

In a previous study we found that N170 responses for words were smaller at left relative to right hemisphere sites in typical readers, but not in dyslexics (Fraga González et al., 2014). Furthermore, we showed the effects of an intervention program with an explicit focus on letter-speech mapping on N170 responses. Finally, N170 amplitude decreased following training in improvers but not in non-improvers (Fraga González et al., 2016).

Objectives

- Evaluate gains on reading fluency after modified 'fluency oriented reading instruction' (FORI, Stahl & Heubach, 2005) in struggling readers
- Explore changes in visual ERPs to words in struggling readers after reading intervention
- Find correlations between neural responses to words and gains in reading fluency measurements.

Methods

Participants:
16 3rd grade children with poor reading fluency skills (<20th percentile); age: M(SD)=8.9(0.4)

Intervention:
Implicit "fluency oriented reading instruction" (FORI).
1-on-1; 45 min sessions
34 sessions; twice a week

ERP experiment
*Block design (**Fig. 1**):*
8 blocks (2 x 2 string types x 2 lengths)
40 trials per block
Trial length: 700ms
Inter-trial interval (ITI): 1350ms *Stimuli:*
Words (CELEX database) and symbol strings (letter-like, **Fig. 2**).
Either short (4-5 characters) or long (6-7 characters).
Task:
Button press when stimuli immediate repetitions are detected (4 per block).

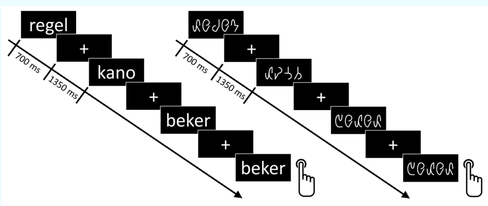

***Figure 1**: Word and symbol string stimuli (examples) visually presented in blocks*

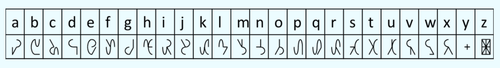

***Figure 2**: False font symbols used to convert words into symbol strings*

ERP analysis:
Biosemi ActiveTwo system
64 scalp electrodes Epoch:(-500 to 1550 ms)
Artifact rejection: Manual and ICA
Reference: average.
Filter: 1-30 Hz
Statistics: Repeated Measures ANOVA
Electrodes in analysis: P9, P7, P5, P10, P8, P6, PO7, PO3, PO8, PO4, O1, O2 (**Fig. 3**)

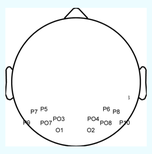

***Figure 3**: Electrodes used in analysis*

Behavioral analysis:
Repeated Measures ANOVA
Improvers vs Poor improvers
(based on the median of the post-pretest difference in the standardized total fluency score for the 3DM word reading task, **Fig. 4**)

Behavioral measurements:
Letter-Speech sound mapping
Word reading (HF,LF,Pseudo)
Spelling
One Minute Test
Text reading

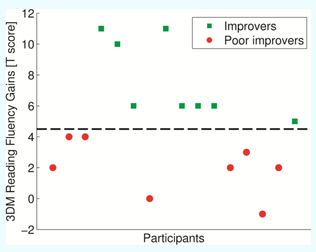

***Figure 4**: Reading Fluency Gains in 3DM word reading task (T scores)*

Results

Struggling readers showed significant group-level increases in age-normed scores of word reading fluency and the speed of letter-speech sound discrimination (**Fig. 5**)

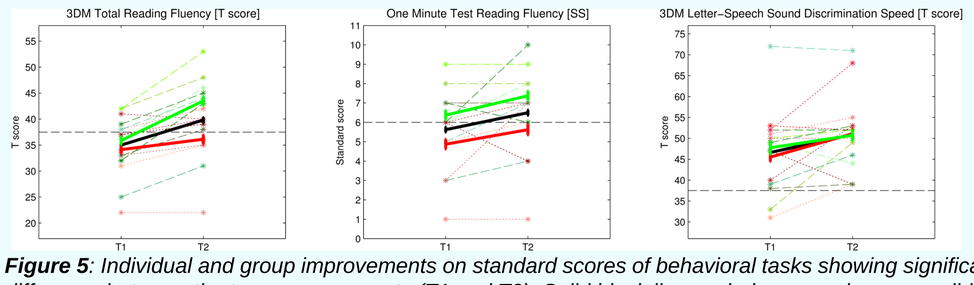

***Figure 5**: Individual and group improvements on standard scores of behavioral tasks showing significant difference between the two measurements (T1 and T2). Solid black line – whole group changes, solid green line – improvers group changes, solid red line – poor improvers group changes; thin green shaded colored dashed lines - improvers, thin red shaded dotted lines – poor improvers; horizontal black line – 10th percentile.*

N170 amplitudes discriminated between words and symbol strings in struggling readers (**Fig. 6**).

N170 responses to words (and symbols) showed no lateralization effects in struggling readers (similar to dyslexic and unlike typical readers in Fraga González et al., 2014).

There was a trend for a differential training effect for words and symbol strings in two hemispheres (Training*StringType*Hemisphere, p=.086).

There were no differences in the age-normed 3DM reading fluency between improvers and poor improvers at the pretest (p=.555), but they differed at the posttest (p<.001). The groups did not differ on any other measure at pre- or posttest.

The training had differential effect on N170 of Improvers and Poor Improvers (Training*Improvement, p=.037). Surprisingly, this was due to the tendency of an overall decrease of N170 amplitude of poor improvers (Training, p =.076; **Fig. 7**)

There was no significant relation between N170 change and gains in reading fluency.

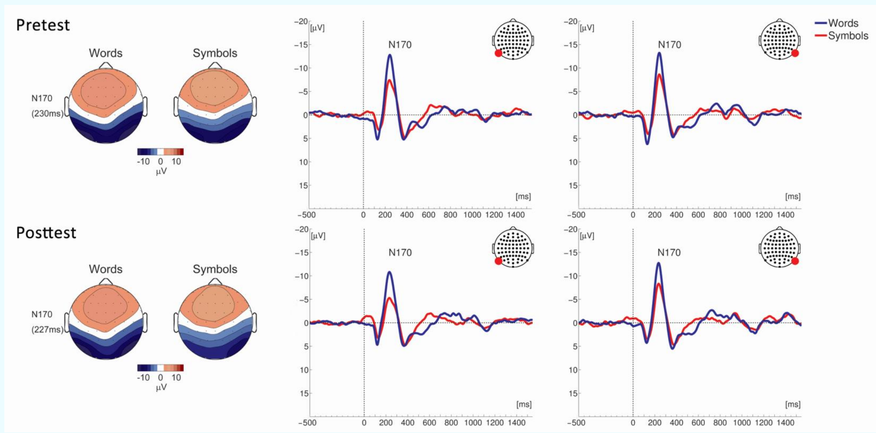

***Figure 6**: Topographical maps showing neural activity fat N170 peak latency (left). Group ERPs for word and symbol stimuli at P9 and P10 (right).*

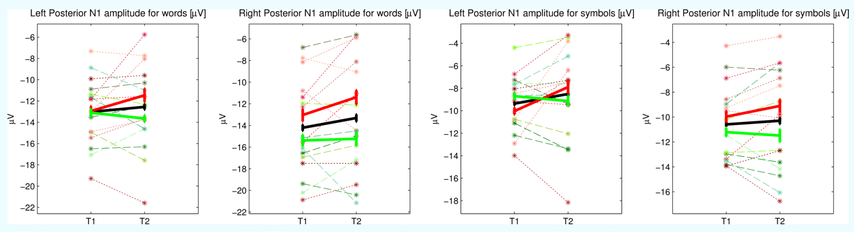

***Figure 7**: Individual and group N170 amplitude (averaged over 6 electrodes) changes between the two measurements (T1 and T2). Solid black line – whole group changes, solid green line – improvers group changes, solid red line – poor improvers group changes, thin green shaded colored dashed lines - improvers; thin red shaded dotted lines – poor improvers.*

Discussion

Intensive and implicit fluency oriented training can improve reading fluency in struggling readers.

Similar to dyslexic readers (Fraga González et al., 2014), struggling readers do not show lateralization effects.

Contrary to the previous study investigating the influence of the explicit training of letter-speech sound mapping (Fraga González et al., 2016), we do not find significant relations between individual behavioral improvements and changes in the N170 amplitude. This may indicate advantage of the explicit reading trainings for initiating both behavioral and neural changes (Fraga González et al., 2015,2016; Fraga González, Žarić et al., 2017; Žarić et al., 2015).

Poor improvers as a group show overall decrease of the N170 amplitude, while improvers N170 amplitude does not change (contrary to Fraga González et al., 2016). This can indicate prolonged stronger reliance on visual encoding in the improvers, while poor improvers may switch to the different strategy as they do not profit from stronger visual encoding. Finally, as current subjects are struggling and not dyslexic readers they may substantially differ in heterogeneity of the impairment.

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